BRK - Hatch

User manual and parameter reference · 1.0.1

Build repeating hat, shaker and ride grooves from a seed, shape the sounds, and keep the take you like. Hatch plays its own synthesis engines and can send the rhythm as MIDI.

Quick start

1. Insert Hatch on an instrument track, choose Trigger: Note, and hold a MIDI note or press PLAY.
2. Choose A–D to hear the four starting grooves. Use NEW for another seed, MUTATE for a smaller change, and Closed/Open/Shaker/Ride to change the layer amounts.
3. Select a sound tab to shape that layer, or select a step to edit it. Save your project or a user preset when you find a take to keep.

Editor window

The editor is 1000 by 740 pixels with HELP closed. Opening HELP adds a footer without moving the existing controls. The footer's Manual / PDF action opens this guide. Shared cells, buttons and the preset bar support keyboard and VoiceOver interaction; drawing and dragging in the step grid require a mouse. Press HELP again to close the help area.

Operation

Note trigger plays while a note is held; Transport trigger follows the host transport. PLAY auditions without a note. The pattern follows the song position, so looping, scrubbing or offline rendering reproduces the same take.

MIDI notes 36–39, labelled C1–D#1 in Hatch, select slots A–D and open the note gate while held. Other notes play the selected slot. Note velocity changes level, and the modulation wheel raises closed-hat Density.

Each slot has its own seed, style, grid, layer amounts and variations. The slots share the cycle length, groove and sound definitions. A cycle can end between bar lines; it repeats at its own length and can drift across the host's bars.

The four slot defaults are the starting grooves. NEW, MUTATE and DICE have a local history reached with the back/forward buttons. A grid paste cannot be undone; clipboard contents and the local take history are not saved with the project.

The five layers are closed hat, open hat, pedal, shaker and ride. Each has two selectable engines and its own sound settings. A closed or pedal hit chokes an open hat; the shaker and ride are independent.

MIDI Out sends General MIDI percussion notes on channel 10: closed hat 42, pedal 44, open hat 46, shaker 70 and ride 51. Host routing support differs by format; route Hatch's MIDI output to another instrument in a host that supports it. This does not mute Hatch's own audio.

HELP follows the hovered or keyboard-focused control. The parameter table documents host parameters; NEW, MUTATE, the dice buttons, PLAY, clipboard actions and hand-edited grid values are described in the operation sections.

Playing it

HELP opens a contextual footer for the hovered or keyboard-focused control; its **Manual / PDF** button opens the full reference at `https://barak007.github.io/brk-audio/manuals/hatch/`. Tab and VoiceOver reach named controls; drawing and dragging in the step grid use a mouse. Opening and closing help preserves the chosen width.

- **Trigger**:
  - **Note**: the pattern plays while any note is held. Where it is in the pattern comes from the host's song position, not from the note, so a note pressed mid-bar comes in mid-pattern and in time. A note up to 30 ms late still plays the hit it just missed, at once. With the transport stopped, the pattern runs from its first step at the note, at the host's tempo (120 without one).
  - **Transport**: plays whenever the host plays.
- **PLAY** (next to Trigger) plays the pattern without a note, the same way a held note does. Use it to audition, or in the standalone app.
- The bars follow the host's bar lines (its last bar start and bar number), also after a change of time signature.
- **C1 to D#1** pick slots A to D (each button shows its note), and also open the gate while held. The last slot picked, by a note or a click on A to D, stays picked after the note ends, and the buttons show it; any other note plays it.
- **Note velocity** 100 and up plays the hits as they are; softer notes play them softer. The **mod wheel** takes the closed hat's Density up toward every step.

The pattern

Four slots, A to D, each a pattern of its own. They start as House, Trap, Garage and Breaks. Every step of each layer gets a score: the style's weight for that place in the beat, blended with the seed's number for that step by **Complexity**. The steps are ranked by score and take hits from the top. Turning an amount up therefore only adds hits, and turning it down only takes the same ones away; a knob never reshuffles the groove.

- **Seed** (0 to 1023): the take. **NEW** picks a random one.
- **Mutation**: each step up re-rolls about an eighth of the steps, a different eighth each time. **MUTATE** takes one step.
- **DICE**, in each section's title (PATTERN, LAYERS, ROLLS, VARIATION, GROOVE, SOUND), rolls that section's controls; **DICE ALL** rolls all of them at once, every sound too. Each control moves from where it is toward a random place in a span that stays musical (a closed hat's decay 25 to 160 ms, Drive up to 9 dB, a ride's engines a ride's), as far as **Range** says: 10% nudges it, 100% lands anywhere in the span. A choice or a seed (Style, Shape, an engine, Decay Seed) changes with Range's chance. A dice never moves the seed, Mutation, Grid, a sound's Level or Pan, or the output. Range is kept with the session, not in presets.
- ‹ and › step back and forth through the slot's NEW, MUTATE and DICE, each giving back exactly the controls it moved (kept while the plugin is open, the last 64).

PATTERN, per slot:

- **Style**: Straight, House, Garage, Techno, Breaks, Trap, or Euclid. Euclid spreads the closed hits as evenly as they go, turned by the seed, so adding a hit spaces them out again.
- **Grid**: 1/8, 1/8 T, 1/16 or 1/16 T steps.
- **Complexity**: 0 plays the style as written; turned up, the seed pulls hits off the obvious places.
- **Evolve**: how much each bar of the cycle differs from the others.
- **Accent**: how much louder the strong places of the beat are than the weak ones.

LAYERS, per slot, in the grid's order: how many hits each row plays.

- **Ride** and **Shaker**: each up to every step (0 at first). The ride leans to quarters or off-beat eighths, the shaker to the sixteenths off the beat, by the style. Neither takes a step from the hats or cuts an open hat.
- **Open** and **Pedal**: each up to half the steps. An open hat takes its step from the closed hat and the pedal.
- **Closed** (its Density): the share of steps the closed hat plays.

ROLLS, per slot:

- **Rolls**: how many closed hits become rolls. Rolls are likelier on the last beat, off the beat, and in the cycle's last bar, and likelier still in Trap and Breaks.
- **Roll Min** and **Roll Max** (2 to 16 hits): how many hits a roll splits its step into; the seed picks each roll's count from Min to Max (the same for a fixed count). A step is the grid's, so the count's speed follows the grid: 2 on a 1/8 grid is a 1/16 roll, on a 1/16 grid a 1/32 roll; 3, 5, 6 and 7 give triplet and odd rolls. A roll fills only its own step, which swing shortens or lengthens.
- **Shape**: how a roll's level moves. Rising, Falling, Flat (all alike after an accented first) or Dip (softest in the middle); Mixed lets the seed pick for each roll. **Depth**: how far it moves, from every hit alike (0) to rising out of near silence (100%).
- **Speed**: a roll's timing. Faster bunches its hits toward the next step, a roll that speeds up; Slower bunches them at its start. A step's own Timing stands in for it.

VARIATION, per slot, each with a seed of its own, so it can be picked again without touching the hits:

- **Decay Vary**: how far each hit's decay moves from its sound's, up to an octave and a half shorter or longer, picked by **Decay Seed** (and moved by Evolve and Mutation like the hits), so a take rings the same every time.
- **Pan Vary**: how far each hit moves from its sound's Pan, either way (0: all in one place). **Pan Drift**: how, from each to a spot of its own (0) to all along one slow curve across the cycle (100). Both picked by **Pan Seed**. Panning only, so mono-safe.

Cycle (shared by the slots, left of the strip under the grid): how long before the pattern comes round again: whole bars, and eighth notes over them, up to 8 bars. It reads "4 bars", "1 bar + 3/8" (a bar and three eighths) or "3/8"; type it the same way, or "1 3/8" or "1.375". Whole bars follow the meter, and the eighths are eighth notes in any meter, so the cut lands on a step (inside one only on a 1/8 T grid). A cycle that ends mid-bar plays its last bar up to there and starts again, so it drifts across the bar lines and meets them again later (1 bar + 3/8 in 4/4, every 11 bars). The grid shades what is past the cut, and the strip draws that bar short.

GROOVE, shared by the slots:

- **Swing**: 0 is straight. Turned up, the second step of each pair comes later: +67% is a triplet shuffle, +100% the hardest (three quarters through the pair). Turned down it comes early, a pushed, rushing feel (-100%:

a quarter through). **Swing On**: Steps swings every second step of the grid; 1/8 swings the eighths, and a 1/16 grid's steps move in proportion (on a 1/8 grid the two are the same). Triplet grids do not swing, nor does a pair the bar cuts short (the last of a 7/16 bar, say). The grid draws the steps where swing puts them.

- **Humanize** plays each hit a touch late (up to 3% of a quarter note) and softer, the same amount each time it plays.

OUTPUT:

- **Dynamics**: how much a hit's velocity changes its level and brightness.
- **Low Cut**: a high pass on everything, 12 dB an octave, from Off to 8 kHz, last in the chain before Output.
- **Output**: the synth's level; Off sends only MIDI.
- **MIDI Out** sends the hits on channel 10 as General MIDI notes: closed F#1 (42), pedal G#1 (44), open A#1 (46), ride D#2 (51), shaker as maracas A#3 (70). An open hat's note ends when the next closed hat or pedal starts, as its sound does; the ride's lasts its decay. Letting go of the note ends every note.

The grid

One bar of the shown slot: one row each for ride, open, closed, pedal and shaker, each step as wide as swing makes it. A hit is as tall as it is loud, split where its roll's hits fall, and each hit shows its ring: how long it rings at the tempo and how it fades (Hold), drawn up to the row's next hit and cut where an open hat is choked. A roll's count sits in its corner. The faint steps are the next ones Density would bring in. A dot marks a step set by hand.

- **Click a row's name** to switch that layer off (its row dims and reads MUTED; it plays no sound and sends no notes, so it chokes nothing either) or on again. Each layer's switch is a parameter too (Closed On, Ride On…).
- **Click** a step to select it; its sound shows in the SOUND tabs. The **STEP** row under the grid shows all its values: Velocity, Shape (its roll rising or falling), Hits, Timing (its roll slowing down or speeding up), Decay, Ring (tight or held), Pan and Sweep. Those the generator chose are dim, those set by hand bright. Drag one (or scroll it) to change it; double-click it to give it back. **RESET** gives back every value of the step. With no step selected, the row shows the step under the mouse (or the last one it was over), dimmer; pressing on it there selects it, and a drag goes on to change it.
- **Selecting holds the bar.** The grid stops following the play: its bar number turns to the accent, a frame goes round it, and the ruler says HELD ON BAR 2. **FOLLOW** (or Esc) follows the play again; so does clicking the held bar in the strip. The arrow keys move the selection; Return adds or takes away the hit.
- **Double-click** a step to add a hit or take it away.
- **Drag** a hit to change two values at once, up and down one and sideways the other (each once it has moved 3 pixels), as the mode next to the strip says:

Mode	Up and down	Sideways
VELOCITY	how loud (the numbers show it)	a roll's level rising (right) or falling
ROLL	how many hits, 1 to 16	a roll speeding up (right) or slowing down

Mode	Up and down	Sideways
DECAY	how long it rings, a quarter to four times its sound's	how it fades: tight (left) or held
PAN	a sweep across (up left to right), drawn as an arrow	where it sits; it then stays put whatever Pan Vary does

A sweep moves a roll's hits along the arrow; a single hit glides along it as it rings (over the first half of its decay).

Each value set by hand stands in for the generator's alone: a step given its own velocity still rolls as Rolls says. **CLEAR** gives back every hand-set step of the slot. Hand-set steps are saved with the session and presets, per slot, bar of the cycle and grid.

Copy, paste and clear are on right-click menus, on the thing itself:

- **a row**: right-click it in the grid (its name or its steps) to copy it, every bar of it (all 8, also those past the cycle), to paste a copied row over it or add it, every bar, or to clear its edits. Any row onto any row: a shaker's row can become the closed hat's, in this slot or another.
- **a bar**: right-click it in the strip under the grid to copy it, every row, to paste a copied bar over it or add it, or to clear its edits.
- **the whole cycle**: right-click any bar in the strip, then *Copy the whole cycle*: the slot as it is, its settings and its hand-set steps. Pasted over another slot, that slot becomes the same, and stays live; added, only its hits (every row, every bar, as they play) come over and the slot keeps its own settings. *Clear the cycle's edits* is CLEAR.

A row or a bar is taken as it plays: every step, the rests as well as the hits, with how loud, its roll (hits, shape and timing), its decay and ring, and a place or sweep set by hand. A bar the cycle cuts short is taken whole. **Paste** puts it in place of what is there, rests and all; **Add** lays only its hits there, so where it rests, what was there stays. The pasted steps become set by hand, so they play as they did wherever they land, and that slot's knobs no longer move them (clearing gives them back to the generator). What stays the slot's own: Pan Vary (each hit still moves about its sound's Pan unless it was placed by hand), Humanize, and each hit's small differences of sound. Since a pasted closed hit is set by hand, an open hat no longer takes its step. A row or a bar, or a cycle added, goes only onto a slot on the grid it was copied on (the menu's item, greyed, says why). One thing is copied at a time, kept while the plugin is open, not saved; a paste cannot be undone.

The strip under the grid shows every bar of the cycle. Click a bar to hold it in the grid; click it again to follow the play; right-click it to copy, paste, add or clear. Faint over each bar, a line per layer whose hits ring for different lengths (Decay Vary, or set by hand): higher for longer, lower for shorter.

The sound

Synthesised live, so every sound control can be automated. Each hit's small differences come from its seed, so a take plays back the same, sample for sample. The closed hat, the open hat, the pedal, the shaker and the ride each have a whole sound of their own, with the same controls. The **SOUND** tabs (**CLOSED**, **OPEN**, **PEDAL**, **SHAKER**, **RIDE**) pick which one shows; selecting a step in the grid shows its row's. **COPY TO** copies the shown sound (Engine A to Attack, and Hold) to another, not its Decay, Level or Pan.

- **Engine A**, **Engine B** and **Morph**, an equal-power crossfade between the two. Every sound can use every engine:
    - **Metal**: six detuned square waves (anti-aliased) into band and high passes. **Color** spreads them apart.
    - **FM**: a carrier and two inharmonic modulators, with feedback. **Color** sets how bright it is.
    - **Modal**: a noise burst into a bank of up to twelve plate-like resonators. **Color** sets how long the high modes ring.
    - **Noise**: filtered noise. **Color** sets how narrow and ringing it is.
    - **Beads**: seeded bead impacts into a small shell's two resonances, a shaker. **Color** sets how many, from a rattle to a hiss; **Pitch** the shell's size.
    - **Bell**: five inharmonic partials, the high ones dying first, a ride's bell. **Color** stretches them from tuned to clangy.
    - **Wash**: twelve detuned high partials, each shimmering slowly, a ride's body. **Color** sets how wide they spread and how much they shimmer.
- **Pitch** (±12 semitones), **Tone** (high and low passes together, darker or brighter), **Snap** (a click of bright noise at the start of each hit), and **Vary** (each hit's own tune, attack, decay and level).
- **Drive**: tanh saturation with an anti-aliased first antiderivative, its input lifted so it keeps the hats' top to about 16 kHz, faded in over its first 3 dB, with its level roughly matched.
- **Attack**: how long each hit takes to come in, from struck (0.4 ms) to swished in (the shaker's 11 ms) and on to 60 ms.
- **Decay**: how long it rings, to -60 dB, times each hit's own (Decay Vary, or set on the grid).
- **Hold**: how the ring fades, ending at the same time. Tight drops fast and leaves a faint tail; Held stays up and then shuts; Natural (the middle) is the plain exponential fade. A step's own Ring adds to it.
- **Level**: how loud it sits against the others. **Pan**: where its hits sit; Pan Vary moves them about it.

The shaker starts as Beads into Noise, the ride as Wash into Bell (Morph between the ride's wash and its bell). **The ride** rings on whatever the hats do. **The pedal** is the foot's chick: a sound of its own (short, dark and quieter to start with), with a short mid knock of the plates meeting. Landing on a ringing open hat, it closes it as a foot does: the ring squeezes shut over 40 ms, its top going first. A closed hat (a stick) cuts it in 25 ms.

Pitch, Attack, the decays, Hold and the Modal, Bell and Wash engines' tuning are set when a hit starts. Morph, Color (except those three's), Tone, Drive, Low Cut and Output move hits that are still ringing too, glided over 20 ms. An open hat's MIDI note lasts its decay (times its own), unless the next closed hat or pedal ends it first.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Trigger	Selects whether held MIDI notes or the host transport open the pattern gate. PLAY can audition either mode.	Note, Transport Default: Note Not host automatable

Parameter	What it does	Values and default
Slot	Selects slot A, B, C or D. Notes 36–39 select the same slots.	A, B, C, D Default: A Not host automatable
Cycle	Sets the repeat length in whole bars and eighth notes, up to eight bars. A partial-bar cycle repeats before the next host bar line.	1/8 to 8 bars Default: 4 bars
Swing	Moves alternate steps earlier or later. Zero is straight; positive values delay them. Triplet grids do not swing.	-100% to +100% Default: Straight
Swing On	Chooses whether swing is based on pairs of grid steps or eighth-note positions.	Steps, 1/8 Default: Steps Not host automatable
Humanize	Adds small, repeatable timing delays and level reductions to individual hits.	0% to 100% Default: 15%
MIDI Out	Enables the generated percussion notes on MIDI channel 10. The internal synth still plays.	Off to On Default: On
Dice Range	Sets how far the section dice move controls from their current positions. This value is saved in the session but excluded from presets.	0% to 100% Default: 30%
Closed On	Enables this layer. A disabled layer produces neither sound nor generated MIDI notes.	Off to On Default: On
Open On	Enables this layer. A disabled layer produces neither sound nor generated MIDI notes.	Off to On Default: On
Pedal On	Enables this layer. A disabled layer produces neither sound nor generated MIDI notes.	Off to On Default: On
Shaker On	Enables this layer. A disabled layer produces neither sound nor generated MIDI notes.	Off to On Default: On
Ride On	Enables this layer. A disabled layer produces neither sound nor generated MIDI notes.	Off to On Default: On
Closed Engine A	Chooses the first synthesis engine for this layer. Every layer can use any engine.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Metal Not host automatable
Closed Engine B	Chooses the second synthesis engine for this layer. Morph blends the two engines.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Noise Not host automatable
Closed Morph	Blends Engine A and Engine B with an equal-power crossfade.	0% to 100% Default: 30%

Parameter	What it does	Values and default
Closed Color	Changes this engine's character: detuning for Metal, brightness for FM, high-mode ring for Modal, filter resonance for Noise, bead density for Beads, partial spacing for Bell and spread/shimmer for Wash.	0% to 100% Default: 50%
Closed Pitch	Transposes the layer's sound in semitones. Tuning is sampled at the start of each hit.	-12.0 st to +12.0 st Default: 0.0 st
Closed Tone	Moves the sound toward darker or brighter filtering. Changes also affect hits already ringing.	Dark 100 to Bright 100 Default: Even
Closed Snap	Adds a short burst of bright noise at the start of each hit.	0% to 100% Default: 30%
Closed Vary	Adds small repeatable differences in each hit's tuning, attack, decay and level.	0% to 100% Default: 35%
Closed Drive	Adds saturation for denser, brighter tails, with approximate level compensation. Changes also affect ringing hits.	0.0 dB to +24.0 dB Default: 0.0 dB
Closed Attack	Sets how quickly each new hit reaches its level.	0.40 ms to 60 ms Default: 0.40 ms
Closed Decay	Sets the layer's nominal decay to -60 dB. Per-hit decay variation and grid overrides scale this time.	15 ms to 400 ms Default: 60 ms
Closed Hold	Shapes the fade within the decay: Tight falls earlier, Natural follows an exponential fade, Held stays up longer before falling.	Tight 100 to Held 100 Default: Natural
Closed Level	Sets this layer's level relative to the other layers.	-24.0 dB to +6.0 dB Default: 0.0 dB
Closed Pan	Sets the layer's stereo position. Per-hit panning variation and grid placement can move or override it.	L 100 to R 100 Default: Centre
Open Engine A	Chooses the first synthesis engine for this layer. Every layer can use any engine.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Metal Not host automatable
Open Engine B	Chooses the second synthesis engine for this layer. Morph blends the two engines.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Noise Not host automatable
Open Morph	Blends Engine A and Engine B with an equal-power crossfade.	0% to 100% Default: 30%

Parameter	What it does	Values and default
Open Color	Changes this engine's character: detuning for Metal, brightness for FM, high-mode ring for Modal, filter resonance for Noise, bead density for Beads, partial spacing for Bell and spread/shimmer for Wash.	0% to 100% Default: 50%
Open Pitch	Transposes the layer's sound in semitones. Tuning is sampled at the start of each hit.	-12.0 st to +12.0 st Default: 0.0 st
Open Tone	Moves the sound toward darker or brighter filtering. Changes also affect hits already ringing.	Dark 100 to Bright 100 Default: Even
Open Snap	Adds a short burst of bright noise at the start of each hit.	0% to 100% Default: 30%
Open Vary	Adds small repeatable differences in each hit's tuning, attack, decay and level.	0% to 100% Default: 35%
Open Drive	Adds saturation for denser, brighter tails, with approximate level compensation. Changes also affect ringing hits.	0.0 dB to +24.0 dB Default: 0.0 dB
Open Attack	Sets how quickly each new hit reaches its level.	0.40 ms to 60 ms Default: 0.40 ms
Open Decay	Sets the layer's nominal decay to -60 dB. Per-hit decay variation and grid overrides scale this time.	80 ms to 2500 ms Default: 450 ms
Open Hold	Shapes the fade within the decay: Tight falls earlier, Natural follows an exponential fade, Held stays up longer before falling.	Tight 100 to Held 100 Default: Natural
Open Level	Sets this layer's level relative to the other layers.	-24.0 dB to +6.0 dB Default: 0.0 dB
Open Pan	Sets the layer's stereo position. Per-hit panning variation and grid placement can move or override it.	L 100 to R 100 Default: Centre
Pedal Engine A	Chooses the first synthesis engine for this layer. Every layer can use any engine.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Metal Not host automatable
Pedal Engine B	Chooses the second synthesis engine for this layer. Morph blends the two engines.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Noise Not host automatable
Pedal Morph	Blends Engine A and Engine B with an equal-power crossfade.	0% to 100% Default: 30%

Parameter	What it does	Values and default
Pedal Color	Changes this engine's character: detuning for Metal, brightness for FM, high-mode ring for Modal, filter resonance for Noise, bead density for Beads, partial spacing for Bell and spread/shimmer for Wash.	0% to 100% Default: 40%
Pedal Pitch	Transposes the layer's sound in semitones. Tuning is sampled at the start of each hit.	-12.0 st to +12.0 st Default: 0.0 st
Pedal Tone	Moves the sound toward darker or brighter filtering. Changes also affect hits already ringing.	Dark 100 to Bright 100 Default: Dark 80
Pedal Snap	Adds a short burst of bright noise at the start of each hit.	0% to 100% Default: 10%
Pedal Vary	Adds small repeatable differences in each hit's tuning, attack, decay and level.	0% to 100% Default: 35%
Pedal Drive	Adds saturation for denser, brighter tails, with approximate level compensation. Changes also affect ringing hits.	0.0 dB to +24.0 dB Default: 0.0 dB
Pedal Attack	Sets how quickly each new hit reaches its level.	0.40 ms to 60 ms Default: 0.40 ms
Pedal Decay	Sets the layer's nominal decay to -60 dB. Per-hit decay variation and grid overrides scale this time.	10 ms to 300 ms Default: 30 ms
Pedal Hold	Shapes the fade within the decay: Tight falls earlier, Natural follows an exponential fade, Held stays up longer before falling.	Tight 100 to Held 100 Default: Natural
Pedal Level	Sets this layer's level relative to the other layers.	-24.0 dB to +6.0 dB Default: -4.4 dB
Pedal Pan	Sets the layer's stereo position. Per-hit panning variation and grid placement can move or override it.	L 100 to R 100 Default: Centre
Shaker Engine A	Chooses the first synthesis engine for this layer. Every layer can use any engine.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Beads Not host automatable
Shaker Engine B	Chooses the second synthesis engine for this layer. Morph blends the two engines.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Noise Not host automatable
Shaker Morph	Blends Engine A and Engine B with an equal-power crossfade.	0% to 100% Default: 25%

Parameter	What it does	Values and default
Shaker Color	Changes this engine's character: detuning for Metal, brightness for FM, high-mode ring for Modal, filter resonance for Noise, bead density for Beads, partial spacing for Bell and spread/shimmer for Wash.	0% to 100% Default: 60%
Shaker Pitch	Transposes the layer's sound in semitones. Tuning is sampled at the start of each hit.	-12.0 st to +12.0 st Default: 0.0 st
Shaker Tone	Moves the sound toward darker or brighter filtering. Changes also affect hits already ringing.	Dark 100 to Bright 100 Default: Bright 20
Shaker Snap	Adds a short burst of bright noise at the start of each hit.	0% to 100% Default: 10%
Shaker Vary	Adds small repeatable differences in each hit's tuning, attack, decay and level.	0% to 100% Default: 50%
Shaker Drive	Adds saturation for denser, brighter tails, with approximate level compensation. Changes also affect ringing hits.	0.0 dB to +24.0 dB Default: 0.0 dB
Shaker Attack	Sets how quickly each new hit reaches its level.	0.40 ms to 60 ms Default: 11 ms
Shaker Decay	Sets the layer's nominal decay to -60 dB. Per-hit decay variation and grid overrides scale this time.	20 ms to 600 ms Default: 90 ms
Shaker Hold	Shapes the fade within the decay: Tight falls earlier, Natural follows an exponential fade, Held stays up longer before falling.	Tight 100 to Held 100 Default: Held 20
Shaker Level	Sets this layer's level relative to the other layers.	-24.0 dB to +6.0 dB Default: -3.1 dB
Shaker Pan	Sets the layer's stereo position. Per-hit panning variation and grid placement can move or override it.	L 100 to R 100 Default: R 30
Ride Engine A	Chooses the first synthesis engine for this layer. Every layer can use any engine.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Wash Not host automatable
Ride Engine B	Chooses the second synthesis engine for this layer. Morph blends the two engines.	Metal, FM, Modal, Noise, Beads, Bell, Wash Default: Bell Not host automatable
Ride Morph	Blends Engine A and Engine B with an equal-power crossfade.	0% to 100% Default: 30%

Parameter	What it does	Values and default
Ride Color	Changes this engine's character: detuning for Metal, brightness for FM, high-mode ring for Modal, filter resonance for Noise, bead density for Beads, partial spacing for Bell and spread/shimmer for Wash.	0% to 100% Default: 50%
Ride Pitch	Transposes the layer's sound in semitones. Tuning is sampled at the start of each hit.	-12.0 st to +12.0 st Default: 0.0 st
Ride Tone	Moves the sound toward darker or brighter filtering. Changes also affect hits already ringing.	Dark 100 to Bright 100 Default: Dark 15
Ride Snap	Adds a short burst of bright noise at the start of each hit.	0% to 100% Default: 15%
Ride Vary	Adds small repeatable differences in each hit's tuning, attack, decay and level.	0% to 100% Default: 30%
Ride Drive	Adds saturation for denser, brighter tails, with approximate level compensation. Changes also affect ringing hits.	0.0 dB to +24.0 dB Default: 0.0 dB
Ride Attack	Sets how quickly each new hit reaches its level.	0.40 ms to 60 ms Default: 0.40 ms
Ride Decay	Sets the layer's nominal decay to -60 dB. Per-hit decay variation and grid overrides scale this time.	300 ms to 6000 ms Default: 1600 ms
Ride Hold	Shapes the fade within the decay: Tight falls earlier, Natural follows an exponential fade, Held stays up longer before falling.	Tight 100 to Held 100 Default: Natural
Ride Level	Sets this layer's level relative to the other layers.	-24.0 dB to +6.0 dB Default: -1.9 dB
Ride Pan	Sets the layer's stereo position. Per-hit panning variation and grid placement can move or override it.	L 100 to R 100 Default: L 25
Dynamics	Sets how strongly hit velocity affects the internal synth's level and brightness.	0% to 100% Default: 70%
Low Cut	High-pass filters the whole synth mix. The lowest setting is off.	Off to 8.00 kHz Default: Off
Output	Sets the internal synth's final output level. It does not scale the outgoing MIDI notes.	Off to +6.0 dB Default: -6.0 dB
Bypass	Bypasses the instrument. This listening state is excluded from presets.	Off to On Default: Off
A Seed	Chooses this slot's repeatable take. The same seed and settings reproduce the same pattern at the same song position.	0 to 1023 Default: 101

Parameter	What it does	Values and default
A Mutation	Changes a small, repeatable share of this slot's steps while keeping the underlying seed.	0 to 255 Default: 0
A Style	Chooses where hits tend to fall. Complexity blends the style's weights with the seeded variation.	Straight, House, Garage, Techno, Breaks, Trap, Euclid Default: House Not host automatable
A Grid	Chooses eighth or sixteenth-note steps, straight or triplet. Hand-edited steps are associated with their grid.	1/8, 1/8 T, 1/16, 1/16 T Default: 1/16 Not host automatable
A Density	Sets how many closed-hat steps play. Increasing it adds hits without reshuffling the existing ones; the mod wheel can raise it further.	0% to 100% Default: 55%
A Complexity	Blends the style's usual hit positions with the seeded variation. Higher values pull hits away from the obvious positions.	0% to 100% Default: 20%
A Evolve	Sets how much the bars within this slot's cycle differ from one another, repeatably.	0% to 100% Default: 25%
A Open	Sets the amount of open hats. An open hat replaces the closed/pedal hit on its step and rings until choked.	0% to 100% Default: 50%
A Pedal	Sets the amount of pedal-hat hits. The pedal has its own sound and closes a ringing open hat.	0% to 100% Default: 0%
A Shaker	Sets how many shaker hits play, up to every step. The shaker has its own sound and does not replace the hats.	0% to 100% Default: 0%
A Ride	Sets how many ride hits play, up to every step. The ride has its own sound and continues independently of hat choking.	0% to 100% Default: 0%
A Accent	Sets the difference in level between strong and weak positions in the pattern.	0% to 100% Default: 50%
A Rolls	Sets how many closed-hat steps become rolls. The style favours different places for rolls.	0% to 100% Default: 10%
A Roll Min	Sets the minimum number of hits within a rolled step. Use the same minimum and maximum for a fixed hit count.	2 hits to 16 hits Default: 2 hits
A Roll Max	Sets the maximum number of hits within a rolled step. The seed chooses the count between the effective limits.	2 hits to 16 hits Default: 4 hits
A Roll Shape	Chooses how hit levels rise or fall within each roll. Mixed lets the seed choose a shape for each roll.	Mixed, Rising, Falling, Flat, Dip Default: Mixed Not host automatable

Parameter	What it does	Values and default
A Roll Depth	Sets how strongly the chosen roll shape changes the hits' levels.	0% to 100% Default: 60%
A Roll Speed	Bends the timing within each rolled step, bunching hits toward its start or end.	Slower 100 to Faster 100 Default: Even
A Decay Vary	Sets how far each hit's decay varies from its sound's decay. The variation is chosen repeatably from Decay Seed.	0% to 100% Default: 0%
A Decay Seed	Chooses the repeatable sequence of per-hit decay changes without changing the pattern's hit positions.	0 to 1023 Default: 101
A Pan Vary	Sets how far hits spread around their sound's Pan position.	0% to 100% Default: 30%
A Pan Drift	Blends separate positions for each hit into a slower panning curve across the cycle.	0% to 100% Default: 0%
A Pan Seed	Chooses the repeatable panning variation without changing the hits.	0 to 1023 Default: 101
B Seed	Chooses this slot's repeatable take. The same seed and settings reproduce the same pattern at the same song position.	0 to 1023 Default: 202
B Mutation	Changes a small, repeatable share of this slot's steps while keeping the underlying seed.	0 to 255 Default: 0
B Style	Chooses where hits tend to fall. Complexity blends the style's weights with the seeded variation.	Straight, House, Garage, Techno, Breaks, Trap, Euclid Default: Trap Not host automatable
B Grid	Chooses eighth or sixteenth-note steps, straight or triplet. Hand-edited steps are associated with their grid.	1/8, 1/8 T, 1/16, 1/16 T Default: 1/16 Not host automatable
B Density	Sets how many closed-hat steps play. Increasing it adds hits without reshuffling the existing ones; the mod wheel can raise it further.	0% to 100% Default: 70%
B Complexity	Blends the style's usual hit positions with the seeded variation. Higher values pull hits away from the obvious positions.	0% to 100% Default: 30%
B Evolve	Sets how much the bars within this slot's cycle differ from one another, repeatably.	0% to 100% Default: 30%
B Open	Sets the amount of open hats. An open hat replaces the closed/pedal hit on its step and rings until choked.	0% to 100% Default: 8%
B Pedal	Sets the amount of pedal-hat hits. The pedal has its own sound and closes a ringing open hat.	0% to 100% Default: 0%

Parameter	What it does	Values and default
B Shaker	Sets how many shaker hits play, up to every step. The shaker has its own sound and does not replace the hats.	0% to 100% Default: 0%
B Ride	Sets how many ride hits play, up to every step. The ride has its own sound and continues independently of hat choking.	0% to 100% Default: 0%
B Accent	Sets the difference in level between strong and weak positions in the pattern.	0% to 100% Default: 55%
B Rolls	Sets how many closed-hat steps become rolls. The style favours different places for rolls.	0% to 100% Default: 40%
B Roll Min	Sets the minimum number of hits within a rolled step. Use the same minimum and maximum for a fixed hit count.	2 hits to 16 hits Default: 2 hits
B Roll Max	Sets the maximum number of hits within a rolled step. The seed chooses the count between the effective limits.	2 hits to 16 hits Default: 6 hits
B Roll Shape	Chooses how hit levels rise or fall within each roll. Mixed lets the seed choose a shape for each roll.	Mixed, Rising, Falling, Flat, Dip Default: Mixed Not host automatable
B Roll Depth	Sets how strongly the chosen roll shape changes the hits' levels.	0% to 100% Default: 60%
B Roll Speed	Bends the timing within each rolled step, bunching hits toward its start or end.	Slower 100 to Faster 100 Default: Faster 40
B Decay Vary	Sets how far each hit's decay varies from its sound's decay. The variation is chosen repeatably from Decay Seed.	0% to 100% Default: 0%
B Decay Seed	Chooses the repeatable sequence of per-hit decay changes without changing the pattern's hit positions.	0 to 1023 Default: 202
B Pan Vary	Sets how far hits spread around their sound's Pan position.	0% to 100% Default: 30%
B Pan Drift	Blends separate positions for each hit into a slower panning curve across the cycle.	0% to 100% Default: 0%
B Pan Seed	Chooses the repeatable panning variation without changing the hits.	0 to 1023 Default: 202
C Seed	Chooses this slot's repeatable take. The same seed and settings reproduce the same pattern at the same song position.	0 to 1023 Default: 303
C Mutation	Changes a small, repeatable share of this slot's steps while keeping the underlying seed.	0 to 255 Default: 0

Parameter	What it does	Values and default
C Style	Chooses where hits tend to fall. Complexity blends the style's weights with the seeded variation.	Straight, House, Garage, Techno, Breaks, Trap, Euclid Default: Garage Not host automatable
C Grid	Chooses eighth or sixteenth-note steps, straight or triplet. Hand-edited steps are associated with their grid.	1/8, 1/8 T, 1/16, 1/16 T Default: 1/16 Not host automatable
C Density	Sets how many closed-hat steps play. Increasing it adds hits without reshuffling the existing ones; the mod wheel can raise it further.	0% to 100% Default: 45%
C Complexity	Blends the style's usual hit positions with the seeded variation. Higher values pull hits away from the obvious positions.	0% to 100% Default: 35%
C Evolve	Sets how much the bars within this slot's cycle differ from one another, repeatably.	0% to 100% Default: 30%
C Open	Sets the amount of open hats. An open hat replaces the closed/pedal hit on its step and rings until choked.	0% to 100% Default: 20%
C Pedal	Sets the amount of pedal-hat hits. The pedal has its own sound and closes a ringing open hat.	0% to 100% Default: 15%
C Shaker	Sets how many shaker hits play, up to every step. The shaker has its own sound and does not replace the hats.	0% to 100% Default: 0%
C Ride	Sets how many ride hits play, up to every step. The ride has its own sound and continues independently of hat choking.	0% to 100% Default: 0%
C Accent	Sets the difference in level between strong and weak positions in the pattern.	0% to 100% Default: 60%
C Rolls	Sets how many closed-hat steps become rolls. The style favours different places for rolls.	0% to 100% Default: 10%
C Roll Min	Sets the minimum number of hits within a rolled step. Use the same minimum and maximum for a fixed hit count.	2 hits to 16 hits Default: 2 hits
C Roll Max	Sets the maximum number of hits within a rolled step. The seed chooses the count between the effective limits.	2 hits to 16 hits Default: 3 hits
C Roll Shape	Chooses how hit levels rise or fall within each roll. Mixed lets the seed choose a shape for each roll.	Mixed, Rising, Falling, Flat, Dip Default: Mixed Not host automatable
C Roll Depth	Sets how strongly the chosen roll shape changes the hits' levels.	0% to 100% Default: 60%

Parameter	What it does	Values and default
C Roll Speed	Bends the timing within each rolled step, bunching hits toward its start or end.	Slower 100 to Faster 100 Default: Even
C Decay Vary	Sets how far each hit's decay varies from its sound's decay. The variation is chosen repeatably from Decay Seed.	0% to 100% Default: 0%
C Decay Seed	Chooses the repeatable sequence of per-hit decay changes without changing the pattern's hit positions.	0 to 1023 Default: 303
C Pan Vary	Sets how far hits spread around their sound's Pan position.	0% to 100% Default: 30%
C Pan Drift	Blends separate positions for each hit into a slower panning curve across the cycle.	0% to 100% Default: 0%
C Pan Seed	Chooses the repeatable panning variation without changing the hits.	0 to 1023 Default: 303
D Seed	Chooses this slot's repeatable take. The same seed and settings reproduce the same pattern at the same song position.	0 to 1023 Default: 404
D Mutation	Changes a small, repeatable share of this slot's steps while keeping the underlying seed.	0 to 255 Default: 0
D Style	Chooses where hits tend to fall. Complexity blends the style's weights with the seeded variation.	Straight, House, Garage, Techno, Breaks, Trap, Euclid Default: Breaks Not host automatable
D Grid	Chooses eighth or sixteenth-note steps, straight or triplet. Hand-edited steps are associated with their grid.	1/8, 1/8 T, 1/16, 1/16 T Default: 1/16 Not host automatable
D Density	Sets how many closed-hat steps play. Increasing it adds hits without reshuffling the existing ones; the mod wheel can raise it further.	0% to 100% Default: 50%
D Complexity	Blends the style's usual hit positions with the seeded variation. Higher values pull hits away from the obvious positions.	0% to 100% Default: 40%
D Evolve	Sets how much the bars within this slot's cycle differ from one another, repeatably.	0% to 100% Default: 40%
D Open	Sets the amount of open hats. An open hat replaces the closed/pedal hit on its step and rings until choked.	0% to 100% Default: 15%
D Pedal	Sets the amount of pedal-hat hits. The pedal has its own sound and closes a ringing open hat.	0% to 100% Default: 10%
D Shaker	Sets how many shaker hits play, up to every step. The shaker has its own sound and does not replace the hats.	0% to 100% Default: 0%

Parameter	What it does	Values and default
D Ride	Sets how many ride hits play, up to every step. The ride has its own sound and continues independently of hat choking.	0% to 100% Default: 0%
D Accent	Sets the difference in level between strong and weak positions in the pattern.	0% to 100% Default: 65%
D Rolls	Sets how many closed-hat steps become rolls. The style favours different places for rolls.	0% to 100% Default: 15%
D Roll Min	Sets the minimum number of hits within a rolled step. Use the same minimum and maximum for a fixed hit count.	2 hits to 16 hits Default: 2 hits
D Roll Max	Sets the maximum number of hits within a rolled step. The seed chooses the count between the effective limits.	2 hits to 16 hits Default: 4 hits
D Roll Shape	Chooses how hit levels rise or fall within each roll. Mixed lets the seed choose a shape for each roll.	Mixed, Rising, Falling, Flat, Dip Default: Mixed Not host automatable
D Roll Depth	Sets how strongly the chosen roll shape changes the hits' levels.	0% to 100% Default: 60%
D Roll Speed	Bends the timing within each rolled step, bunching hits toward its start or end.	Slower 100 to Faster 100 Default: Even
D Decay Vary	Sets how far each hit's decay varies from its sound's decay. The variation is chosen repeatably from Decay Seed.	0% to 100% Default: 0%
D Decay Seed	Chooses the repeatable sequence of per-hit decay changes without changing the pattern's hit positions.	0 to 1023 Default: 404
D Pan Vary	Sets how far hits spread around their sound's Pan position.	0% to 100% Default: 30%
D Pan Drift	Blends separate positions for each hit into a slower panning curve across the cycle.	0% to 100% Default: 0%
D Pan Seed	Chooses the repeatable panning variation without changing the hits.	0 to 1023 Default: 404

Presets and files

The host session saves the pattern, hand-edited steps, sound parameters and selected slot/preset. User presets also save the four slots, sounds and step overrides, with stable choice IDs.

User presets use .hatch files in ~/Library/Audio/Presets/Barakigal Audio/Hatch. There is no factory preset bank; the four initial slots are the starting points. Loading a preset leaves Bypass and Dice Range unchanged.

The copy/paste clipboard and the last 64 NEW/MUTATE/DICE history entries exist only while the plugin is open. A paste cannot be undone; save a preset or project before making a change you may want to keep separately.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.